Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General data

1. General

Provision for TCAS II is made and can be installed as customer order option 6219.

A CAS 67A TCAS II provides both Traffic Advisories (TA) and Resolution Advisories (RA) to the flight crew for appropriate vertical avoidance maneuvers when a possible collision is threatened with an intruding aircraft.

2. Description

A. Traffic Alert and Collision Avoidance System II (TCAS II)

The CAS 67A TCAS II includes all the functionality of TCAS I, but also includes pitch targets (avoidance and fly-to zones) and aural advisories, known as Resolution Advisories (RAs). There are two types of RAs:

1 Corrective – Pilot action is required. Corrective RAs instruct the pilot to take avoidance action such as, "DESCEND, DESCEND", "CLIMB, CLIMB", or "ADJUST VERTICAL SPEED".

2 Preventative – Current path and speed must be maintained. Preventative RAs warn the pilot not to deviate from the current path and speed by instructing, "MONITOR VERTICAL SPEED" or "MAINTAIN VERTICAL SPEED".

TCAS II systems coordinate the resolution advisories between aircraft before issuing commands to the pilots, so that when one aircraft is instructed to descent, the other is typically told to climb. Traffic symbols including intruder symbol, data tag, and vertical speed arrow are consistent between the Multi-Function Display (MFD) and Horizontal Situation Indicator (HSI) overlay. The number of intruders displayed on the HSI overlay is limited to 32.

Intruder numbers assigned by the TCAS processor are used to associate intruder parameters with the correct intruders. A pattern of intruders, as transmitted by the TCAS unit, is displayed on the traffic overlay, when the TCAS is in test mode and traffic is selected for display. When a TCAS resolution advisory is displayed, the TCAS pitch command is displayed on the Attitude Director Indicator (ADI). The red goal posts indicate the do-not-fly zone (or zones, if both pitch-up and pitch-down avoidance zones are displayed). The green box indicates the fly-to zone. When the aircraft symbol is inside the red do-not-fly zones, it turns red. If the aircraft symbol is inside the green fly-to box, it turns green. Otherwise, it remains white. Traffic has display priority over weather (WX) radar and flight plan symbols (Terrain has priority over TCAS).

Generated TCAS II processor audio signals are sent to the audio isolation amplifier (refer to 23-54-00, Audio System Isolation Amplifier – General data) and then to the unmuted audio control panel line. The TCAS voice messages are suppressed when the EGPWS sends voice messages to the audio control panel. Traffic Advisories (TA) and Resolution Advisories (RA) from the TCAS II processor trigger pulse light operation when the landing light switch is selected to the upper TCAS position.

The CAS 67A TCAS II processor unit and configuration module (CM 2000) are mounted in the nose avionics rack. A directional antenna mounted on the top of the fuselage at station FS110 is used to determine intruder bearing. A omni-directional antenna is mounted on the bottom of the fuselage at station FS67.4. During TCAS transmission on the directional or omni-directional antenna, the TCAS generates a suppression pulse to inhibit other L-band installed equipment transmission.

28 Volt DC power is protected by a 5 Amp TCAS (H5) (Left Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The TCAS II interfaces to the following aircraft systems:

1) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

2) Enhanced Ground Proximity Warning System (EGPWS) (refer to 34-44-00, Enhance Ground Proximity Warning System (TAWS) (Mod 6/2049) – General Data).

3) Audio Isolation Amplifier (refer to 23-54-00, Audio System Isolation Amplifier – General data).

4) Lights (refer to 33-00-00, Lights – General Data).

5) A suppression pulse to/from the Dual KXP Transponder system (refer to 34-53-00, KXP 2290 Transponder System (Mod 6/2049) – General Data).

6) A suppression pulse to/from the Distance Measuring Equipment (DME) (refer to 34-54-00, Distance Measuring Equipment (DME) (Mod 6/2049) – General Data).

3. Adjustment/Test

A. Function Test TCAS II (CAS 67A)

1) General

These procedures are for the function test of the TCAS II (CAS 67A).

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that TCAS (H5) circuit breaker is closed.

d) Initiate a TCAS self test in the XPDR/TCAS detail page.

e) Verify the following:

- A white TEST caption is displayed on the lower left corner of the PFDs.
- Resolution Advisory (solid red square) will appear at 3 o'clock, range of 2 nm, at 200 feet relative altitude, above (+02) with no VS arrow.
- Traffic Advisory (solid yellow circle) will appear at 9 o'clock, range of 2 miles, at 200 feet relative altitude, below (–02) with ascending arrow.
- Proximity traffic (solid white diamond, or Honeywell – Cyan) will appear at 1 o'clock, range 3.6 miles, at 1000 feet relative altitude, below (–10) with descending arrow.
- Non-Threat traffic (open white diamond, or Honeywell – Cyan) will appear at 11 o'clock, range of 3.6 miles, at 1000 feet relative altitude, above (+10) with no arrow.

f) Confirm that “TCAS SYSTEM TEST OK” is audible after the test. Should a failure be detected during self test, the audio message says, “TCAS SYSTEM TEST FAIL”

g) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Electrical Bonding Test – Upper TCAS II Antenna (ANT 67A)

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to the cockpit Temperature Sensor MTG. Over Head Console panel, in the flight compartment.

3) Remove screws and open panel.

4) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the TCAS Upper Antenna mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

5) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

6) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

7) Close Temperature Sensor MTG. Over Head Console Panel and tighten the retaining screws.

C. Electrical Bonding Test – Lower TCAS II Antenna (CL–110–41–30)

1) Ensure the electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to left-hand Access and Inspection Provision panel 15. (Refer to 12-00-00, Servicing – General Data).

3) Turn DZUS fasteners 1/4 turn and swing open access panel.

4) Locate TCAS Lower Antenna forward of station 73.0.

5) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the TCAS Lower Antenna mounting screw head. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

6) Check the electrical conductivity between several random mounting screws within the electrical within the electrical bond contact area to ensure consistency.

7) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

8) Close left-hand Access and Inspection Provision 15 panel and tighten DZUS fasteners.

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND LIGHTNING. ENSURE INTEGRITY OF THE HIRF/LIGHTNING PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – PANEL / PEDESTAL MOUNT UNITS CANNOT BE RELIABLY GROUNDED TO THE AIRFRAME VIA THE PANEL MOUNTING SCREWS OR DZUS FASTENERS. GROUNDING OF THESE UNITS IS ACCOMPLISHED BY USING THE SHIELD GROUNDING BRAID OR JUMPER. VERIFY PROPER ELECTRICAL BONDING AS DETAILED IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE CONNECTOR AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of the TCAS II (CAS 67A)

1) General

These procedures are for the removal and installation of the TCAS II (CAS 67A).

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the TCAS (H5) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the forward avionics shelves through the forward baggage compartment.

d) Loosen the two retaining screw clamps from the front and slide the unit out of the mounting tray.

e) Visually inspect mounting tray, if damaged replace as per Para 4.B.

f) Remove unit from aircraft.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the TCAS (H5) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Slide the unit into the mounting tray.

d) Tighten the two retaining screw clamps at the front lip of the TCAS unit.

e) Reconnect the electrical connector to the unit. Ensure connections are secure.

f) Install top and middle covers attached to bulkhead station 37.21.

WARNING **ACCESS COVERS ARE REQUIRED TO MAINTAIN PROTECTION FEATURES THAT ARE ESSENTIAL TO MINIMIZING ADVERSE EFFECTS FROM HIGH INTENSITY RADIATED FIELDS (HIRF).**

g) Remove safety clip and close TCAS (H5) circuit breaker on the avionics circuit breaker panel.

h) Test the TCAS system as per Para 3.A.2).

B. Removal and Installation TCAS II Mounting Tray

1) General

These procedures are for the removal and installation of the TCAS II mounting tray.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Remove the CAS 67A TCAS II unit as per Para 4.A.2).

c) Remove screws securing the mounting tray to the instrument panel.

d) Remove mounting tray from aircraft.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Align the tray to the mounting holes on the instrument panel.

c) Install the screws that secure the mounting tray to the front avionics rack.

d) Install the CAS 67A TCAS II unit as per Para 4.A.3).

e) Test the TCAS II system as per Para 3.A.2).

C. Removal and Installation Configuration Module (CM 2000)

1) Removal

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Pull the TCAS (H5) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the configuration module which is mounted in the forward avionics compartment (underside of middle shelf).

d) Disconnect the electrical connectors from the unit.

e) Remove the two screws that secure the configuration module.

f) Remove configuration module from aircraft.

Note Stow connectors of the removed components.

2) Installation

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the TCAS (H5) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Gain access to the configuration module which is mounted in the forward avionics compartment (underside of middle shelf).

d) Reconnect the electrical connector to the configuration module. Ensure connections are secure.

e) Remove safety clip and close TCAS (H5) circuit breaker on the avionics circuit breaker panel.

f) Verify the configuration module setup.

D. Removal and Installation of the Top Directional Antenna (ANT 67A)

1) General

These procedures are for the removal and installation of the top directional TCAS II antenna (ANT 67A).

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the TCAS (H5) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Carefully cut the sealant from around the base of the antenna.

d) Remove the four screws from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connections.

f) Before disconnecting the antenna cables, ensure that the four antenna cables are labelled correctly and correspond to the correct post on the antenna.

g) Disconnect the coaxial connectors.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the TCAS (H5) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Solvent clean the electrical contact bond areas immediately before the installation (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage.

e) Ensure that the four antenna cables are labelled correctly and correspond to the correct post on the antenna.

f) Connect the coaxial connectors.

g) Secure the antenna assembly on fuselage with four mounting screws.

h) Solvent clean the exposed area of the airframe around the antenna (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

i) Touch up the exposed area of airframe around the antenna if required with primer.

j) If the affected area requires paint touch up, paint the area with the top-coat paint system matching the existing.

k) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data). The aircraft will be ready to fly approximately 24 hours after sealing.

l) Check the antenna electrical bonding as per Para 3.B.

m) Remove safety clip and close TCAS (H5) circuit breaker on the avionics circuit breaker panel.

n) Do a function test on the TCAS system as per Para 3.A.2).

E. Removal and Installation of the Bottom Omni-Directional Antenna (CL–110–41–30)

1) General

These procedures are for the removal and installation of the bottom omni-directional (CL–110–41–30) TCAS II antenna.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the TCAS (H5) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Carefully cut the sealant from around the base of the antenna.

d) Remove the four screws from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connectors.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the TCAS (H5) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Solvent clean the electrical contact bond areas immediately before the installation. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with four mounting screws.

f) Solvent clean the exposed area of the airframe around the antenna. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data). The aircraft will be ready to fly approximately 24 hours after sealing.

j) Check the antenna electrical bonding as per Para 3.C.

k) Remove safety clip and close TCAS (H5) circuit breaker on the avionics circuit breaker panel.

l) Do a function test on the TCAS system as per Para 3.A.2).

5. Troubleshooting

A. TCAS II (CAS 67A)

1) If the TCAS II (CAS 67A) is deemed inoperative check the following:

a) Ensure TCAS (H5) circuit breaker is closed.

b) Check that the units are securely installed and the locking screw is securely tightened.

c) Visually inspect TCAS unit connectors for damaged wires or contacts.

d) Ensure that the DME (refer to 34-54-00, Distance Measuring Equipment (DME) (Mod 6/2049) – General Data) and Transponder (refer to 34-53-00, KXP 2290 Transponder System (Mod 6/2049) – General Data) suppression lines are functioning correctly.

e) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

f) Ensure that the antenna coaxial cable connector are properly mated and secure.

g) Inspect antenna external surfaces for damage.

h) Check that the antenna is properly mounted and screws are tight.

i) If repair or service is required, send to an appropriate rated avionics repair facility.

Note The TCAS II (CAS 67A) contains BITE (Built-In Test Equipment) to enable the operational health of the unit to be constantly monitored. When a critical fault is detected, the CAS 67A will notify the Apex maintenance system, which will record the fault. The CAS 67A also stores detected failures in non volatile memory for later review.

Note If appropriately rated for avionics repair, shop fault isolation procedures are contained in the CAS 67A CMM.